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For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

PDR 400

ETHAMBUTOL TABLETS BP 400 mg

Composition
Each film coated tablet contains :
Ethambutol Hydrochloride BP 400 mg

Product Description:
PDR 400 (Ethambutol Tablets BP 400mg) is white circular, biconvex, film coated tablets.

Pharmacodynamics:
Pharmacotherapeutic group: Antimycobacterial, ATC code: J04AK02.
Mode of Action
Ethambutol is bacteriostatic. It is effective against Mycobacterium tuberculosis and M. bovis with an MIC of 0.5 - 8 µg per ml. the exact mechanism of action is unknown.
While it has activity against some atypical Mycobacteria including M. kansasii, activity against other micro-organisms has not yet been reported.
It is effective against tubercle bacilli resistant to other tuberculostatics.
Mechanism of Resistance
Cross-resistance has not yet been reported. Primary resistance to ethambutol is uncommon but resistant strains of M. tuberculosis are readily produced if ethambutol is used alone.

Pharmacokinetics:
Ethambutol is readily absorbed after oral administration and this absorption is not significantly impaired by food. After a single dose of 25mg/kg body weight, within 4 hours peak plasma concentrations of up to 5µg/ml are obtained, by 24 hours the concentration decreases to less than 1µg/ml. Most of a dose is excreted unchanged in the urine and up to 20% in faeces, within 48 hours. From 8 - 15% of a dose appears in urine as inactive metabolites.
Ethambutol readily diffuses into red blood cells and into the cerebrospinal fluid when the meninges are inflamed. It has also been reported to cross the placenta.

Indication:
Ethambutol Tablets is used in combination with other antituberculosis agents for the treatment of all forms of tuberculosis caused by Mycobacterium tuberculosis.
Ethambutol Tablets 400mg is also used in the treatment of infections caused by atypical mycobacteria, such as Mycobacterium avium complex (MAC).

Recommended Dose:
Initial treatment (patients never on antituberculosis therapy): 15 mg/kg body weight as a single oral dose once every 24 hours.
In the more recent studies, isoniazid has been administered concurrently in a single, daily, oral dose.
Retreatment (patients previously on antituberculosis therapy):
25 mg/kg body weight as a single oral dose once every 24 hours.
Concurrently administer at least one other antituberculous drug to which the organisms have been demonstrated to be susceptible.
It is not recommended for use in patients below 13 years old.

Route of Administration: ORAL.

Contraindication:
Ethambutol is contraindicated in patients who are known to be hypersensitive to the drug. It is also contraindicated in patients with known optic neuritis and poor vision unless clinical judgement determines that it may be used.

Warning and Precautions:
Renal function:
Toxic effects are more common if renal function is impaired.
Ocular toxicity:

Ethambutol may produce a unique type of visual impairment which is generally reversible and which appears to be due to optic neuritis and to be related to dose and duration of treatment.
Less than 1% of patients undergoing treatment with the higher dose regimen of 25mg/kg/day for two months, and 15mg/kg/day thereafter, have exhibited decrease in visual acuity. It is recommended that patients undergo a full ophthalmic examination before starting treatment. This should include visual acuity, colour vision, perimetry and ophthalmoscopy. Any change may be unilateral or bilateral and hence both eyes should be tested individually.
Routine ophthalmological examination for adults is not thereafter necessary, but patients should be informed the importance of reporting any change in vision. Routine ophthalmological examinations may be considered desirable when treating young children.
Any negative effects on vision are generally reversible when administration of the drug is discontinued promptly and recovery of visual acuity has usually occurred over a period of weeks to months after the drug was discontinued. Patients have then received Ethambutol at lower dosages without toxicity.
In rare cases, recovery may be delayed for up to one year or more or the effects may be irreversible.

Hepatic impairment:
Liver function tests should be performed in patients who develop symptoms suggestive of hepatitis or who become generally unwell during treatment.
Other warnings:
Consideration should be given to current clinical guidance on the appropriate use of antituberculous drugs.

Interaction of Other Medicaments:
Doses of uricosurics may need to be increased, since ethambutol competes with uric acid for its renal excretion.
Concomitant therapy with disulfiram may increase the risk for ocular toxicity.
Disulfiram is a drug used to support the treatment of chronic alcoholism by producing an acute sensitivity to alcohol. Disulfiram, a chelating agent, may produce a toxic optic neuropathy.
Both Ethambutol and Disulfiram are having the neurotoxic side effects, and the Concomitant administration of the Ethambutol with other neurotoxic medications (Disulfiram) may increase the risk of developing the neurotoxicity.
Aluminium hydroxide impairs the absorption of ethambutol. During therapy with ethambutol acid-suppressing drugs or antacids not containing aluminium hydroxide should be used during Ethambutol therapy.

Pregnancy and Lactation:
Pregnancy
The potential risk for humans is unknown.
Ethambutol is not recommended during pregnancy and in women of childbearing potential unless the potential benefit to the mother is considered to outweigh any possible risks.
Breast-feeding
Ethambutol/metabolites have been identified in breastfed newborns/infants of treated women.
Breast-feeding is not recommended during Ethambutol treatment unless the benefit of breast-feeding to the child is considered to outweigh any possible risks.

Side Effects:
In this section, frequencies of undesirable effects are defined as follows: Very common (≤1/10); common (≤1/100, < 1/10); uncommon (1/1,000, < 1/100); rare (≤1/10,000, < 1/1,000); very rare (< 1/10,000); not known (frequency cannot be estimated from the available data)

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Blood and lymphatic system disorders:	thrombocytopenia	rare
	leukopenia, neutropeniaeosinophilia	very rare
Immune system disorders	hypersensitivity, anaphylactoid reactions allergic reactions, anaphylaxis, allergic pneumonitis.	very rare
Metabolism and nutrition disorders	hyperuricaemia.	uncommon
	gout.	very rare
Psychiatric disorders	mental confusion, hallucination	very rare
Nervous system disorders	peripheral neuritis, peripheral neuropathy, paraesthesia of the extremities, numbness	rare
	burning pain, weakness (hands and feet), dizziness, headache, disorientation	very rare
	tremor	unknown
Eye disorders	optic neuritis, retrobulbar neuritis*(decreased visual acuity, loss of vision, scotoma, colour blindness, visual disturbance, visual field defect, eye pain)	Common
Respiratory, thoracic and mediastinal disorders	pneumonitis, pulmonary infiltrates, with or without eosinophilia	very rare
Gastrointestinal disorders	nausea, vomiting, anorexia, abdominal pain & diarrhoea have been noted in patients on multiple drug anti- tuberculosis therapy including ethambutol although not in test patients receiving ethambutol as sole therapy.	not known
	Flatulence, metallic taste, loss of appetite, upset stomach	
'Hepatobiliary disorders:	Hepatic reactions with hepatitis, jaundice, transient increase in liver enzymes, abnormal liver function test values and very rarely hepatic failure have been reported in patients treated with multiple drug therapy including ethambutol. Liver function tests should be performed in patients who develop symptoms suggestive of hepatitis or who become generally unwell during treatment.	not known
Musculoskeletal and connective tissue disorders	joint pains	very rare
Skin & subcutaneous Tissue disorders	Rash, pruritus, urticaria.	Rare
	photosensitive lichenoid eruptions, bullous dermatitis, Stevens Johnson syndrome, epidermal necrolysis.	very rare
Renal and urinary disorders	nephrotoxicity including interstitial nephritis.	very rare
General disorders and administration site conditions	malaise, pyrexia.	very rare

* This effect is thought to be dose related, and frequency is dependent on both dose and duration of treatment. It occurs most frequently with doses of 25 mg/kg body weight and after two months of therapy, however optic neuritis has also occurred after only a few days of therapy. The effect is often reversible upon discontinuation of therapy. To avoid permanent damage visual acuity should be checked regularly during treatment and therapy discontinued immediately when visual disturbances occur.
Visual disturbances may be unilateral or bilateral; therefore each eye should be tested separately. Typical signs include: blurred vision, eye pain, impairment of colour vision (red-green colour blindness), constriction of visual field (central or peripheral scotoma), and any loss in vision. Recovery of visual acuity has usually occurred over a period of weeks to months after the drug was discontinued, and patients have then received Ethambutol at lower dosage without toxicity.
¹Liver function tests should be performed in patients who develop symptoms suggestive of hepatitis or who become generally unwell during treatment.

Symptoms and Treatment of Overdose:
Symptoms: Gastrointestinal disturbances, vomiting, fever, headache, anorexia, dizziness, hallucinations and/or visual disturbances.
Treatment: There is no specific antidote, but gastric lavage should be employed if necessary.

Effects on Ability to Drive and Use Machine:
Ethambutol may produce a unique type of visual impairment, Dizziness, disorientation, numbness and paraesthesia of the extremities has been reported. Therefore, patients should be advised not to drive or operate machinery if they experience any of these effects.

Instructions for Use:
Patients should be instructed to swallow the whole tablet.

Presentation:
Blister pack of 24 x 28 Tablets, Blister pack of 10 x 10 Tablets or Blister pack of 50 x 10 Tablets.

Storage:
"Store in the original container".
Store in a cool and dry place, below 30°C, protected from light.
Keep out of reach and sight of children.

Shelf Life: 60 months.

Manufactured by :
Svizera Labs Pvt. Ltd.
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